

- a) The set of integers between 1 and 50 divisible by 8.
- b) the set $S = \{x|x^2 + 4x - 5 = 0\}$
- c) the set $S = \{x|x \text{ is a continent}\}$

Answer of exercise 25

- a) $\{8, 16, 24, 32, 40, 48\}$
- b) Solving the quadratic equation gives: $x^2 + 4x - 5 = 0 \Leftrightarrow (x + 2)^2 - 9 = 0 \Leftrightarrow x + 2 = \pm 3 \Leftrightarrow x = 1 \vee x = -5$. Thus, $S = \{x|x^2 + 4x - 5 = 0\} = \{-5, 1\}$
- c) $S = \{x|x \text{ is a continent}\} = \{\text{Africa, Europe, Asia, North-America, South-America, Antarctica, Australia}\}$

Exercise 26

Describe the sample space S consisting of all points in the first quadrant inside a circle of radius 3 with center at the origin.

Answer of exercise 26

Recall that an equation for a circle with radius r is $x^2 + y^2 = r^2$. The circle of interest can be described by $x^2 + y^2 = 9$. Since we want the points inside the region inside the circle in the first quadrant, we obtain $S = \{x|x^2 + y^2 < 9; x \geq 0; y \geq 0\}$

Exercise 27

An experiment involves tossing a pair of dice, one green and one red, and recording the numbers that come up. If x equals the outcome on the green die and y the outcome on the red die, describe the sample space S .

- a) In terms of elements using the notation (x, y)
- b) list the elements corresponding to the event A that the sum is greater than 8.
- c) list the elements corresponding to the event B that a 2 occurs on either die.
- d) list the elements corresponding to the event C that a number greater than 4 occurs on the green die.
- e) list the elements corresponding to the event $A \cap C$
- f) list the elements corresponding to the event $A \cap B$
- g) list the elements corresponding to the event $B \cap C$

Answer of exercise 27

- a) $S = \{(x, y)|x, y \in \{1, 2, 3, 4, 5, 6\}\} = \{(1, 1), (1, 2), \dots, (6, 6)\}$
- b) $\{(3, 6), (4, 5), (4, 6), (5, 4), (5, 5), (5, 6), (6, 3), (6, 4), (6, 5), (6, 6)\}$
- c) $\{(2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (1, 2), (3, 2), (4, 2), (5, 2), (6, 2)\}$
- d) $\{(x, y)|x \in \{5, 6\}, y \in \{1, 2, 3, 4, 5, 6\}\}$
- e) $A \cap C = \{(5, 4), (5, 5), (5, 6), (6, 3), (6, 4), (6, 5), (6, 6)\}$
- f) $A \cap B = \emptyset$
- g) $B \cap C = \{(5, 2), (6, 2)\}$

Exercise 28

Two jurors are selected from 4 alternates to serve at a murder trial. Using the notation $A_1 A_3$ for example, to denote the simple event that alternates 1 and 3 are selected, list the 6 elements of the sample space S .

Answer of exercise 28

$A_1A_2; A_1A_3; A_1A_4; A_2A_3; A_2A_4; A_3A_4$

Exercise 29

Consider the sample space $S = \{ \text{copper, sodium, nitrogen, potassium, uranium, oxygen, zinc} \}$ and the events

$A = \{ \text{copper, sodium, zinc} \}$

$B = \{ \text{sodium, nitrogen, potassium} \}$

$C = \{ \text{oxygen} \}$

List the elements of the sets corresponding to the following events:

- a) A'
- b) $A \cup C$
- c) $(A \cap B') \cup C'$
- d) $B' \cap C'$
- e) $A \cap B \cap C$
- f) $(A' \cup B') \cap (A' \cap C)$

Answer of exercise 29

- a) $A' = \{ \text{nitrogen, potassium, uranium, oxygen} \}$
- b) $A \cup C = \{ \text{copper, sodium, zinc, oxygen} \}$
- c) $(A \cap B') \cup C' = \{ \text{copper, sodium, nitrogen, potassium, uranium, zinc} \}$
- d) $B' \cap C' = \{ \text{copper, uranium, zinc} \}$
- e) $A \cap B \cap C = \emptyset$
- f) $(A' \cup B') \cap (A' \cap C) = \{ \text{copper, nitrogen, potassium, uranium, oxygen, zinc} \} \cap \{ \text{oxygen} \} = \{ \text{oxygen} \}$

Exercise 30

If $S = \{x|0 < x < 12, x \in \mathbb{R}\}$, $M = \{x|1 < x < 9, x \in \mathbb{R}\}$ and $N = \{x|0 < x < 5, x \in \mathbb{R}\}$, find

- a) $M \cup N$
- b) $M \cap N$
- c) $M' \cap N'$

Answer of exercise 30

This question can best be answered by drawing two number lines and specifying the sets on them. The solutions are:

- a) $M \cup N = \{x|0 < x < 9, x \in \mathbb{R}\}$
- b) $M \cap N = \{x|1 < x < 5, x \in \mathbb{R}\}$
- c) $M' \cap N' = \{x|9 \leq x < 12, x \in \mathbb{R}\}$

Exercise 31

If each coded item in a catalog begins with 3 distinct letters followed by 4 distinct nonzero digits, find the probability of randomly selecting one of these coded items with the first letter a vowel and the last digit being even.

Answer of exercise 31

There are 5 vowels and 4 even non-zero digits, hence: $\frac{5}{26} \times \frac{4}{9} = \frac{10}{117}$.

Exercise 32

An automobile manufacturer is concerned about a possible recall of its best-selling four-door sedan. If there were a recall, there is a probability of 0.25 of a defect in the brake system, 0.18 of a defect in the transmission, 0.17 of a defect in the fuel system, and 0.40 of a defect in some other area.

- a) What is the probability that the defect is the brakes or the fueling system if the probability of defects in both systems simultaneously is 0.15?
- b) What is the probability that there are no defects in either the brakes or the fueling system?

Answer of exercise 32

Let A denote the event that there is a defect in the brake system, and B the event that there is a defect in the fuel system. Given is the following:

- $P(A) = 0.25$
 - $P(B) = 0.17$
 - $P(A \cap B) = 0.15$
- a) Using the equation $P(A \cup B) = P(A) + P(B) - P(A \cap B)$, we obtain $P(A \cup B) = 0.25 + 0.17 - 0.15 = 0.27$
 - b) Using $A' \cap B' = (A \cup B)'$ and $P(A) = 1 - P(A')$, we get $P(A' \cap B') = 1 - 0.27 = 0.73$

Exercise 33

If a letter is chosen at random from the English alphabet, find the probability that the letter

- a) is a vowel exclusive of y
- b) is listed somewhere ahead of the letter j
- c) is listed somewhere after the letter g

Answer of exercise 33

- a) $\frac{5}{26}$
- b) $\frac{9}{26}$
- c) $\frac{19}{26}$

Exercise 34

A pair of fair dice is tossed. Find the probability of getting

- a) a total of 8
- b) at most a total of 5

Answer of exercise 34

- a) $\frac{5}{36}$
- b) $\frac{5}{18}$

Exercise 35

In a poker hand consisting of 5 cards, find the probability of holding

- a) 3 aces
b) 4 hearts and 1 club.

Answer of exercise 35

- a) A deck of cards contains 52 cards. 4 of these are aces, hence 48 are not. There are $\binom{52}{5}$ ways to pick a hand of 5 cards. There are $\binom{4}{3}$ ways to pick three aces, and $\binom{48}{2}$ ways to pick two cards that are not aces. It follows that the probability of interest is $\frac{\binom{4}{3}\binom{48}{2}}{\binom{52}{5}}$
- b) There are 13 hearts and 13 clubs. Applying the same reasoning as in question (a), we obtain $\frac{\binom{13}{4}\binom{13}{1}}{\binom{52}{5}}$

Exercise 36

If three books are picked at random from a shelf containing 5 novels, 3 books of poems, and a dictionary, what is the probability that

- a) the dictionary is selected?
b) 2 novels and 1 book of poems are selected?

Answer of exercise 36

To solve this question, we apply the same method as in Exercise 35. (Note that we choose three out of nine books.)

$$\text{a) } \frac{\overbrace{\binom{1}{1}}^{\text{Dictionary}} \overbrace{\binom{8}{2}}^{\text{other}}}{\underbrace{\binom{9}{3}}_{\text{total}}}$$

$$\text{b) } \frac{\binom{5}{2}\binom{3}{1}}{\binom{9}{3}}$$

Exercise 37

In a high school graduating class of 100 students, 54 students studied mathematics, 69 students studied history, and 35 studied both mathematics and history. If one of these students is selected at random, find the probability that

- a) the student took mathematics or history?
b) the student did not take either of these subjects?
c) the student took history but not mathematics?

Answer of exercise 37

Let M denote the event that a student took mathematics and H the event that (s)he took history. The following is given:

- $P(M) = 0.54$
- $P(H) = 0.69$
- $P(M \cap H) = 0.35$

It may be convenient to visualize the information using a Venn-diagram.

- a) Using $P(M \cup H) = P(M) + P(H) - P(M \cap H)$, we get $P(M \cup H) = 0.54 + 0.69 - 0.35 = 0.88$.
- b) Using $A' \cap B' = (A \cup B)'$ and $P(A') = 1 - P(A)$, we obtain $P(M' \cap H') = 1 - 0.88 = 0.12$.
- c) The solution is $P(H) - P(M \cap H) = 0.69 - 0.35 = 0.34$.